SUPSI

BBC News Classification

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Project Overview

The aim of this project is to **predict the category** to which BBC articles belong, by comparing various NLP approaches and analysing key hyperparameters.

Task 1

Web Scraping and Dataset

Task 2

Rule Based Approach

Task 3

Classification algorithms

Task 1: Scraping and Datasets

FIRST VERSION

- **2000** articles x **3** columns
- **Classes**: 1000 sport + 1000 non-sport
- **Split**: 0.8 train + 0.2 test

SECOND VERSION

- **7049** articles x **4** columns (url is unused)
- **Classes**: some are unbalanced (F1, golf, ...)
- **Split**: 0.8 train + 0.2 test

PRE - PROCESSING

Use of a **regex-based tokenizer**, removing stopwords and non-alphabetic tokens.

Task 2: Rule Based Approach (2000 articles)

KEYWORD RATIO

```
Keyword SPORT selezionate: 271
Keyword NON-SPORT selezionate: 1241
```

```
--- Top 20 keyword SPORT ---
liv        ratio=1.000  sport=   63  non=    0
celtic     ratio=1.000  sport=   56  non=    0
leagues    ratio=1.000  sport=   35  non=    0
durde      ratio=1.000  sport=   51  non=    0
bulls      ratio=1.000  sport=   38  non=    0
```

Use a keyword report to identify articles with promising results.

THRESHOLDS

```
SOGLIA_FREQ_MIN = 30
SOGLIA_SPORT = 0.85
SOGLIA_NON_SPORT = 0.15
```

A minimum threshold ensures accurate results for the majority classes.

RESULTS

	NS	S
NS	198	2
S	15	185

F1 = 0.96

uses defined linguistic rules that capture grammar, syntax, and vocabulary, often as “if–then” patterns, to analyze or generate text, without learning from data.

Task 3: Logistic Regression

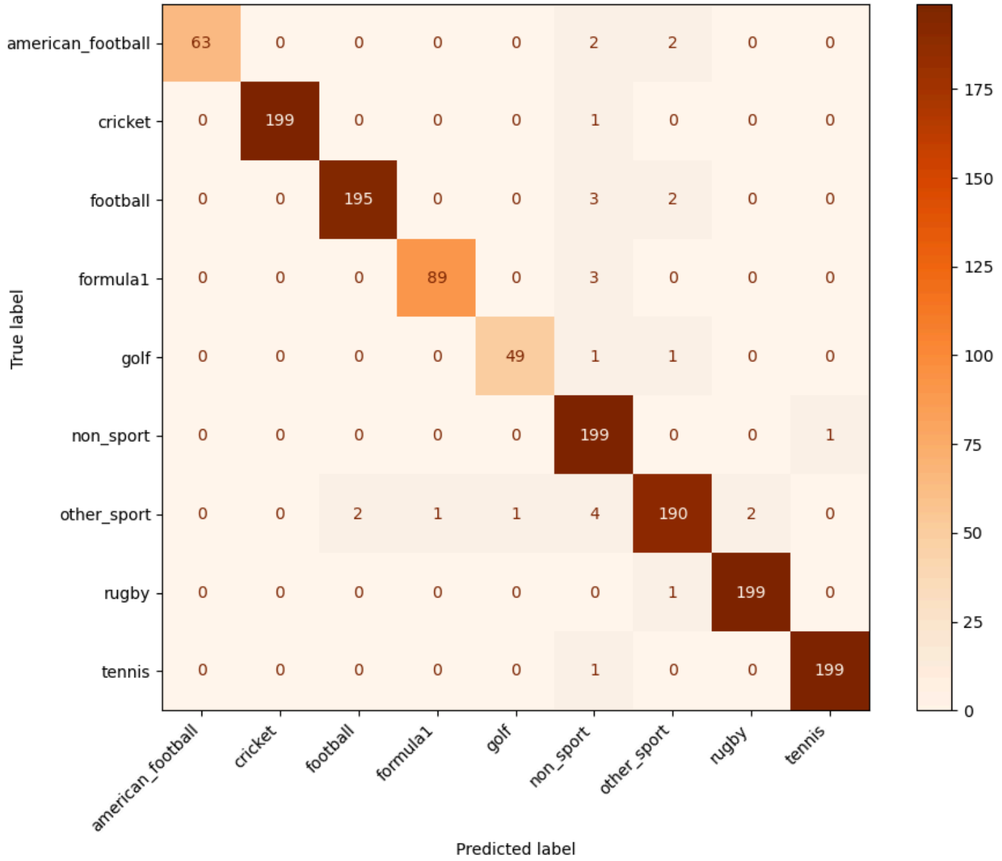
Performance

```
=== Classification Report — Logistic Regression ===
                   precision    recall  f1-score   support

american_football       1.00      0.94      0.97        67
          cricket       1.00      0.99      1.00       200
         football       0.99      0.97      0.98       200
         formula1       0.99      0.97      0.98        92
             golf       0.98      0.96      0.97        51
        non_sport       0.93      0.99      0.96       200
      other_sport       0.97      0.95      0.96       200
            rugby       0.99      0.99      0.99       200
           tennis       0.99      0.99      0.99       200

         accuracy                           0.98      1410
        macro avg       0.98      0.97      0.98      1410
     weighted avg       0.98      0.98      0.98      1410
```

Confusion Matrix



Task 3: NN Feedforward

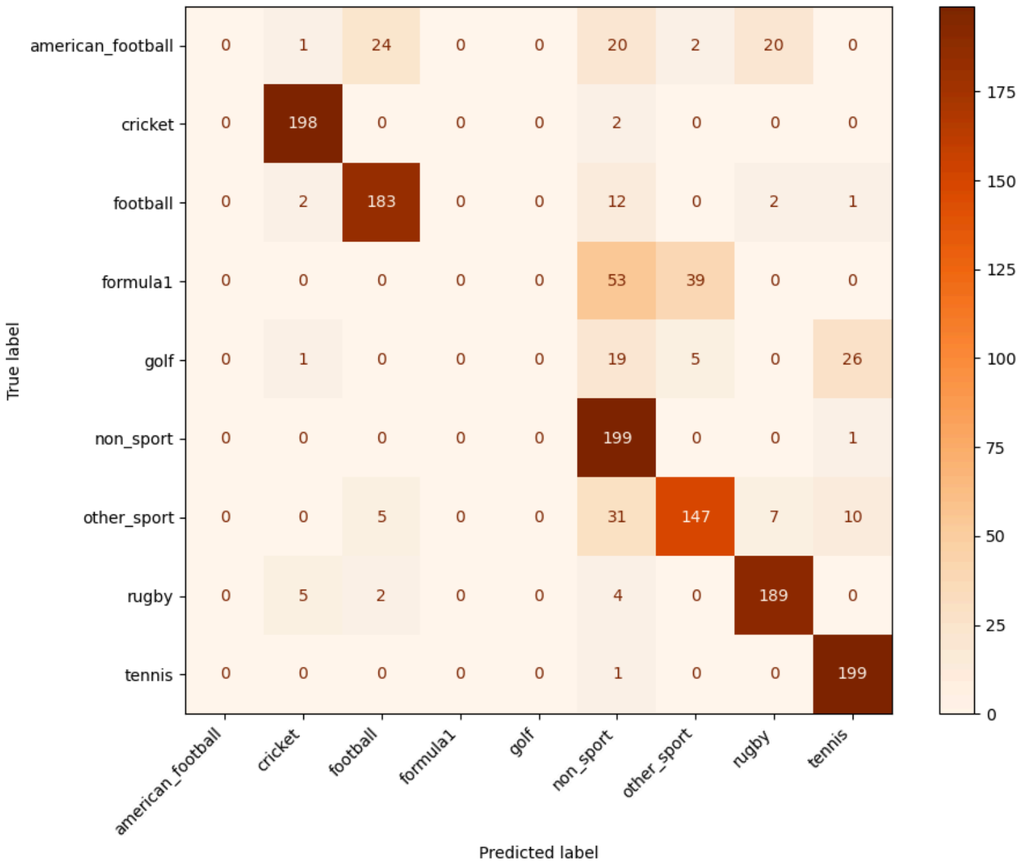
Performance

```
=== Classification Report — NN Feedforward ===
                  precision    recall  f1-score   support

american_football       0.00      0.00      0.00        67
          cricket       0.96      0.99      0.97       200
         football       0.86      0.92      0.88       200
         formula1       0.00      0.00      0.00        92
             golf       0.00      0.00      0.00        51
        non_sport       0.58      0.99      0.74       200
      other_sport       0.76      0.73      0.75       200
            rugby       0.87      0.94      0.90       200
           tennis       0.84      0.99      0.91       200

         accuracy                           0.79      1410
        macro avg       0.54      0.62      0.57      1410
     weighted avg       0.69      0.79      0.73      1410
```

Confusion Matrix



Task 3: NN Feedforward - Adam + EarlyStopping

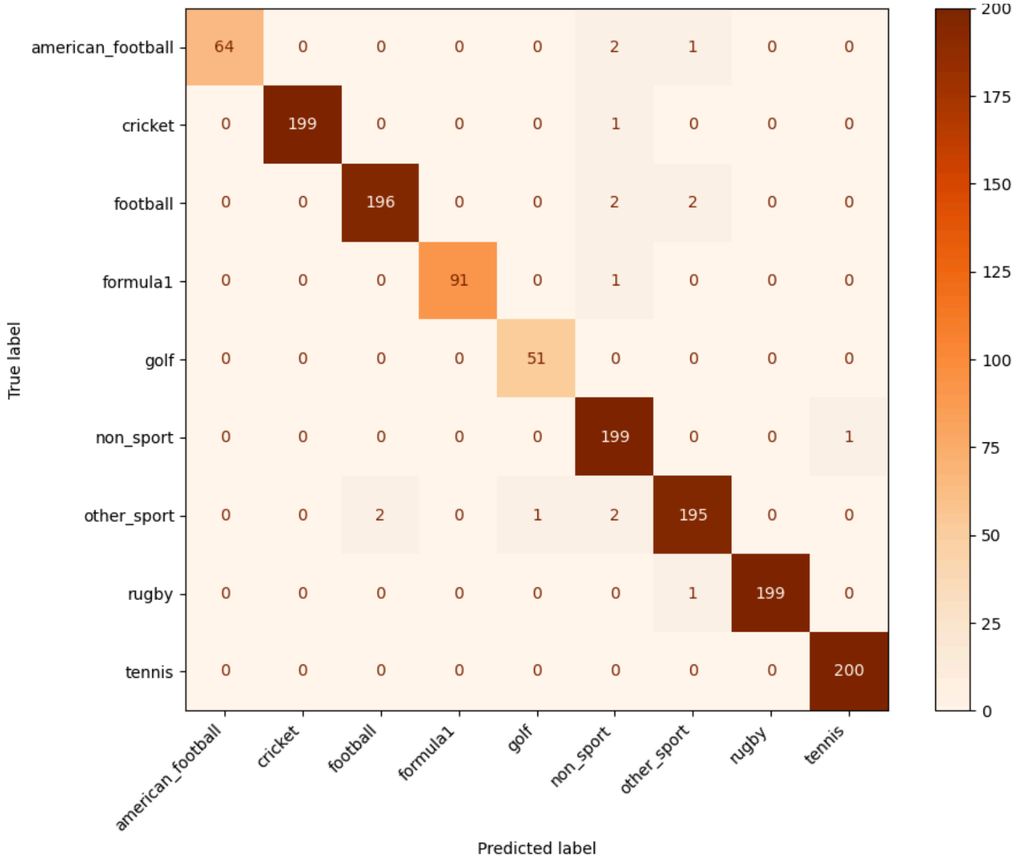
Performance

```
=== Classification Report - NN Feedforward (Adam+EarlyStoppin
                   precision    recall  f1-score   support

american_football       1.00      0.96      0.98        67
          cricket       1.00      0.99      1.00       200
         football       0.99      0.98      0.98       200
         formula1       1.00      0.99      0.99        92
             golf       0.98      1.00      0.99        51
        non_sport       0.96      0.99      0.98       200
      other_sport       0.98      0.97      0.98       200
            rugby       1.00      0.99      1.00       200
           tennis       1.00      1.00      1.00       200

         accuracy                           0.99      1410
        macro avg       0.99      0.99      0.99      1410
     weighted avg       0.99      0.99      0.99      1410
```

Confusion Matrix



Task 3: Word2Vec (GloVe) - IDF

Performance

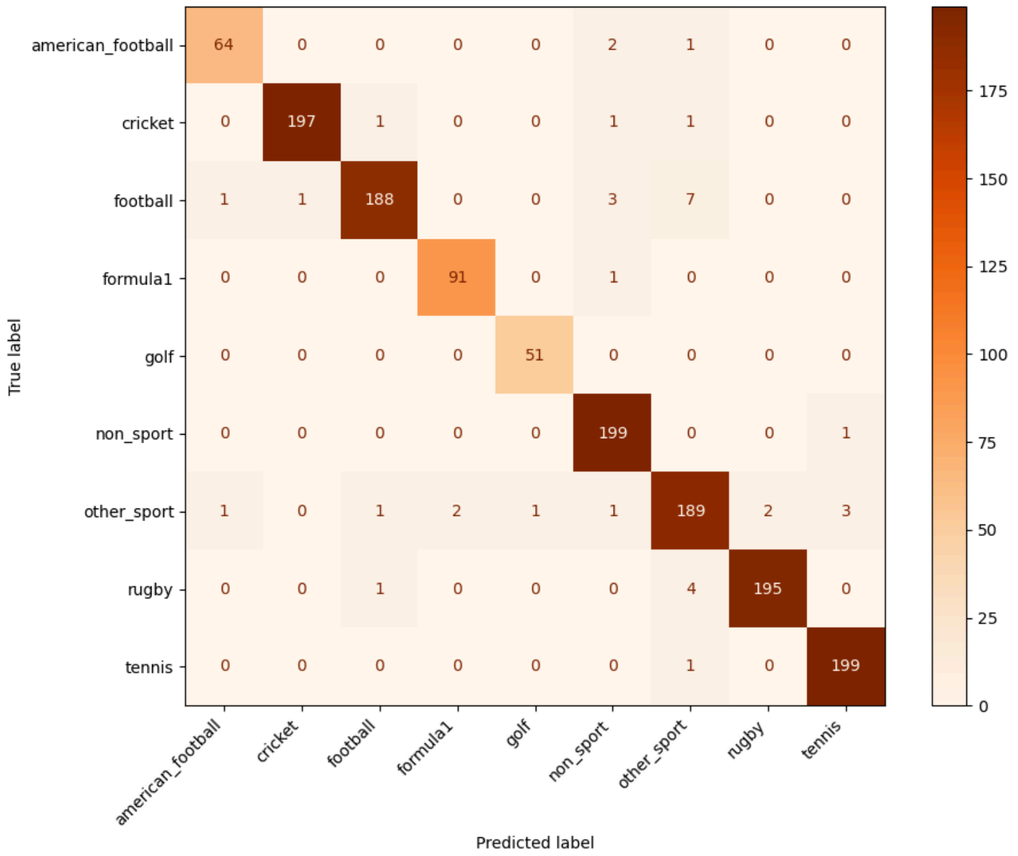
```
=== Classification Report — Word2Vec + FFN ===
                   precision    recall  f1-score   support

american_football       0.97      0.96      0.96        67
          cricket       0.99      0.98      0.99       200
         football       0.98      0.94      0.96       200
         formula1       0.98      0.99      0.98        92
             golf       0.98      1.00      0.99        51
        non_sport       0.96      0.99      0.98       200
         accuracy                           0.97      1410
        macro avg       0.97      0.98      0.97      1410
     weighted avg       0.97      0.97      0.97      1410
```

```
Parole vicine a 'goal':
  goals                0.836
  scored               0.676
  scoring              0.659
  kick                 0.598
  minute               0.589
```

```
Parole vicine a 'football':
  soccer               0.768
  basketball           0.734
  league               0.660
  baseball             0.648
  rugby                0.643
```

Confusion Matrix



Task 3: Sentence Transformer - Chunked

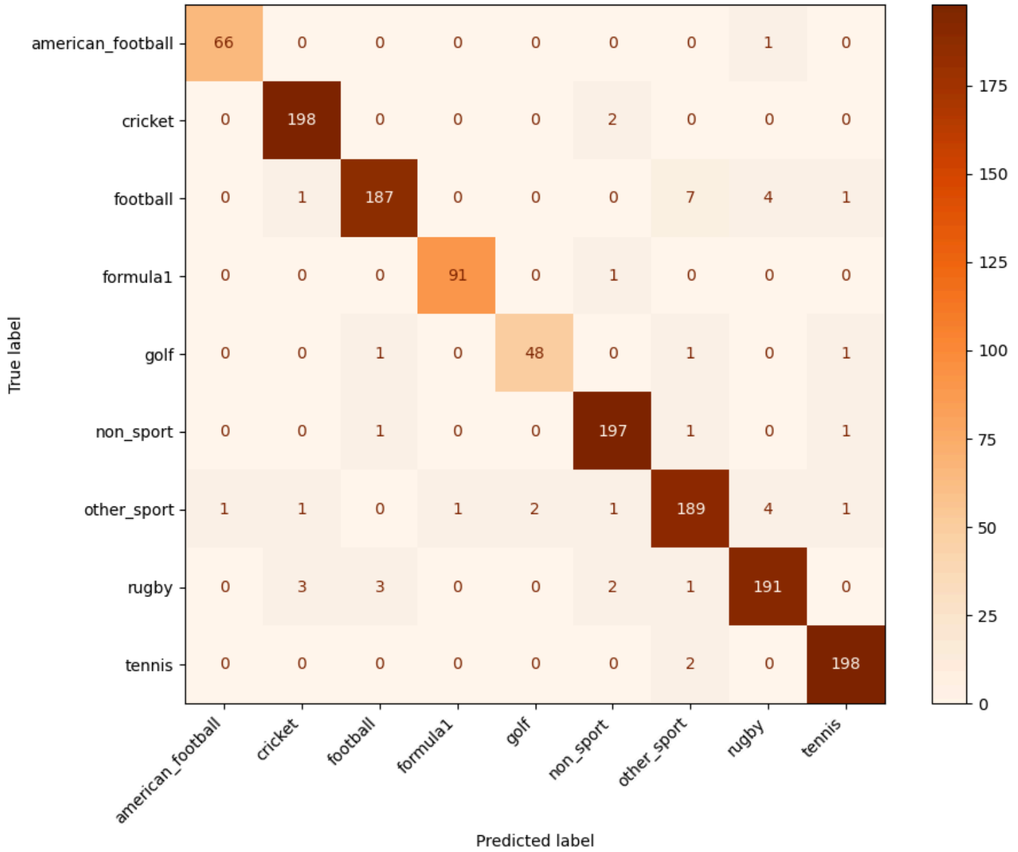
Performance

```
=== Classification Report — Sentence Transformer + FFN ===
                  precision    recall  f1-score   support

american_football       0.99      0.99      0.99        67
          cricket       0.98      0.99      0.98       200
         football       0.97      0.94      0.95       200
         formula1       0.99      0.99      0.99        92
             golf       0.96      0.94      0.95        51
        non_sport       0.97      0.98      0.98       200
      other_sport       0.94      0.94      0.94       200
            rugby       0.95      0.95      0.95       200
           tennis       0.98      0.99      0.99       200

         accuracy                           0.97      1410
        macro avg       0.97      0.97      0.97      1410
     weighted avg       0.97      0.97      0.97      1410
```

Confusion Matrix



Task 3: BERT fine-tuned

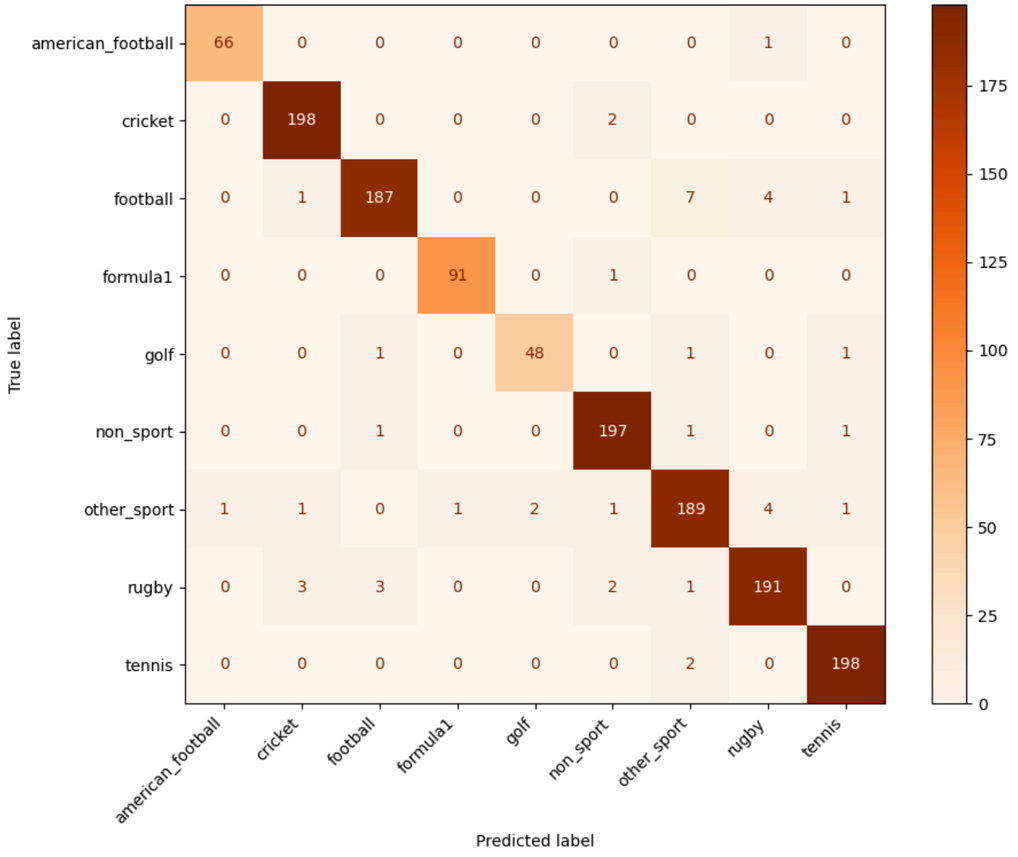
Performance

```
=== Classification Report — BERT (DistilBERT fine-tuned) ===
                  precision    recall  f1-score   support

american_football     0.9565    0.9851    0.9706        67
          cricket     1.0000    0.9950    0.9975       200
         football     0.9797    0.9650    0.9723       200
         formula1     0.9892    1.0000    0.9946        92
             golf     0.9623    1.0000    0.9808        51
        non_sport     0.9898    0.9700    0.9798       200
      other_sport     0.9740    0.9350    0.9541       200
            rugby     0.9614    0.9950    0.9779       200
           tennis     0.9804    1.0000    0.9901       200

         accuracy                         0.9794      1410
        macro avg     0.9770    0.9828    0.9797      1410
     weighted avg     0.9796    0.9794    0.9793      1410
```

Confusion Matrix



Task 3: stratified 5-fold CV and recap

Logistic Regression

- **LR CV accuracy**: 0.9823 ± 0.0022
- **LR CV F1 macro**: 0.9809 ± 0.0032

NN Feedforward + Adam

- **NN CV accuracy**: 0.9898 ± 0.0017
- **NN CV F1 macro**: 0.9905 ± 0.0028

Metodo	Accuracy	F1 macro
Baseline Rule-Based	0.9241	0.9275
Logistic Regression (TF-IDF)	0.9801	0.9784
NN Feedforward – SGD (TF-IDF)	0.7908	0.5729
NN Feedforward – Adam (TF-IDF)	0.9887	0.9883
Word2Vec (GloVe) + FFN	0.9745	0.9751
Sentence Transformer + FFN	0.9688	0.9702
BERT (DistilBERT fine-tuned)	0.9794	0.9797

Thank You

LET’S RUN THE **STREAMLIT DEMO** :)